Statistical Data Analyst: Longitudinal Data Exercise

Instructions

Please complete these series of data management and analysis exercise. Please submit both your .rmd file, merged dataset and the HTML output to the questions below.

Description

This dataset comes from an observation longitudinal cohort study. The aim of the study is to calculate incident hypertension and to understand the risk factors associated with incident cases. Participants were asked to come back for their annual visit with opportunity to come back for unscheduled visits if they needed clinical care (unofficial visits). Blood pressure, weight and medication use was measured at all clinical visits. Community health workers made home visits every 6 months and collected blood pressure data. Deaths were ascertained using phone calls and participants were able to exit the study at any time.

The following datasets are attached:

1. baseline_df.csv contains the entry visit data.
2. visits_df.csv contains the blood pressure data at each visit.
3. outcomes_df.csv contains data on any study exits or deaths.

Variables:

- id: unique participant ID
- age: participant age in years at baseline
- sex: participant sex (1 = male, 2 = female)
- height: participant height in cm
- visit_date: date of visit
- visit_type: Categorization of visit types.
    - Scheduled visits are labelled by year of follow-up in months (i.e. 1 year follow-up = M12)
    - Unscheduled visits are labelled as ‘unofficial’
    - Home visits are labelled as ‘home’.
- sbp1: first systolic blood pressure (SBP) reading
- sbp2: second SBP reading
- sbp3: third SBP reading
- dbp1: first diastolic blood pressure (DBP) reading
- dbp2: second DBP reading
- dbp3: third DBP reading
- weight: participant weight in kilograms
- amlodipine: reported use of calcium-channel blocker
- bisoprolol: reported use of beta-blocker
- captopril: reported use of ACE inhibitors
- carvedilol: reported use of beta-blocker
- enalapril: reported use of ACE inhibitors
- furosemide: reported use of loop diuretics

- hctz: reported use of thiazide diuretics
- lisinopril: reported use of ACE inhibitors
- losaratan: reported use of ARBs
- metoprolol: reported use of beta blockers
- aspirin: reported use of aspirin
- atorvastatin: reported use of statins
- simvastatin: reported use of statins
- metformin: reported use of diabetes medication
- glyburide: reported use of diabetes medication
- free_text_med: Free text used by clinicians to put medication use

Antihypertensive medication use follows under use of any of these medication categories:

- Thiazide
- Calcium channel blockers
- ARBs
- ACE inhibitors
- Beta Blockers
- Loop diuretics
- Mineralocorticoid Receptor Antagonists (MRAs)

Your task is to merge the datasets together to 1) calculate the incidence of hypertension per 1,000 person-years and 2) to identify the risk factors associated with incident hypertension.

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Part 1 – Data cleaning

A. Merge the datasets together to create one long dataset. There may be data quality issues and ambiguities—handle them as you see appropriate.
B. Use the 2nd and 3rd blood pressure readings to create an average reading. Use this to calculate incident hypertension. Use up to last date seen (either clinic or home visit) as the end of follow-up.
C. Save this dataset.

Note: Keep all code you used in your file output including the ones you may have used to visualise or clean data.

Part 2 – Incident hypertension

	Disease Free (n)	**Events (n)**	**Person-years**	**Incidence per 1,000 PYs**
Total				
Sex				
Male				
Female				
Age group				
18-29 years				
30-39 years				
40-49 years				
50-59 years				
60+ years				

A. Create a table output with the following information (see table above) using the definition of hypertension as <u>first</u> clinic visit with ≥130/80 mmHg or antihypertensive medication use.

B. Create a table output with the following information (see table above) using the definition of hypertension as the <u>second</u> clinic or home visit (does not need to be consecutive) with ≥130/80 mmHg or antihypertensive medication use.

Part 3 – Regression

Conduct univariable and multivariable regression to identify baseline risk factors for incident hypertension using the definition of hypertension as <u>first</u> clinic visit with ≥130/80 mmHg or antihypertensive medication use. Adjust for age (by 10-year increments), sex and baseline BMI categories.